

Biology Practice Test 02 (Adaptive)

Format: ISC Subjective

Portion: Set 1 Midterm

Time Allowed: 2 Hours

Maximum Marks: 60

General Instructions:

- All questions are compulsory. Internal choice is provided in Section D.
 - Maintain neat handwriting and use a pencil for diagrams.
 - **NO BLANKS ALLOWED:** A -1 mark penalty will be applied to your final score for any question left completely blank. (Brain Dump technique).
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SECTION A (10 Marks)

Answer all questions. Each question carries 1 mark.

Question 1: A person is rushed to the hospital after stepping on a rusty nail and is immediately injected with anti-tetanus serum. This is an example of: (a) Naturally acquired active immunity (b) Artificially acquired active immunity (c) Artificially acquired passive immunity (d) Naturally acquired passive immunity

Question 2: The Law of Independent Assortment is physically explained by the random alignment of homologous chromosome pairs during which phase? (a) Prophase I (b) Metaphase I (c) Anaphase II (d) Telophase I

Question 3: Which phase of the logistic population growth curve corresponds to the carrying capacity (K) of the environment?

Question 4: Expand the acronym ZIFT in the context of Assisted Reproductive Technologies.

Question 5: What does a high Biochemical Oxygen Demand (BOD) indicate about a water body?

Question 6: Which of the following is not an outbreeding device in angiosperms? (a) Dichogamy (b) Cleistogamy (c) Self-incompatibility (d) Herkogamy

Question 7: Name the embryonic stage that implants into the uterine endometrium in humans.

Question 8: The drug 'Smack' is obtained from the latex of which plant?

Question 9: State the Hardy-Weinberg algebraic equation for a population in genetic equilibrium.

Question 10: Name the specific bacterium primarily responsible for the large holes in 'Swiss cheese'.

SECTION B (14 Marks)

Answer all questions. Each question carries 2 marks.

Question 11: Define *Apomixis* strictly in botanical terms using exact NCERT/ISC phrasing.

Question 12: State two fundamental differences between divergent and convergent evolution.

Question 13: Mention two significant uses of the fungus *Trichoderma* in agriculture.

Question 14: Differentiate between a test cross and a back cross.

Question 15: Give two reasons why Medical Termination of Pregnancy (MTP) is heavily regulated by law in India.

Question 16: Explain the difference between normal cells and cancerous cells with respect to "contact inhibition".

Question 17: Mention two structural adaptations seen in desert plants (xerophytes) to minimize water loss.

SECTION C (21 Marks)

Answer all questions. Each question carries 3 marks.

Question 18: Explain the specific roles of GnRH, Luteinizing Hormone (LH), and Follicle Stimulating Hormone (FSH) in regulating spermatogenesis in human males.

Question 19: Draw a neat, properly labeled diagram showing the structure of a mature human sperm.

Question 20: Explain the mechanism of sex determination in birds, highlighting how it differs from humans.

Question 21: How do homologous organs provide evidence for evolution? Explain this concept using the example of the forelimbs of mammals.

Question 22: A couple is unable to conceive due to oligospermia in the male partner. Suggest and briefly explain an ART technique most suitable for this specific condition.

Question 23: Briefly describe the Miller-Urey experiment. What did this experiment conclusively prove regarding the origin of life?

Question 24: Explain the concept of incomplete dominance using the flower colour of *Antirrhinum* (Snapdragon) as an example. Show the cross.

SECTION D (15 Marks)

Answer any ONE question from each of the three pairs below.

Question 25: (Option A) (a) Describe the famous experiment conducted by Matthew Meselson and Franklin Stahl using heavy nitrogen isotopes. (4 Marks) (b) What was the final, definitive conclusion of their experiment? (1 Mark) **OR (Option B)** (a) Describe the detailed structure of a polynucleotide chain of DNA. (3 Marks) (b) State the Central Dogma of Molecular Biology and name the enzyme responsible for reversing this flow of information. (2 Marks)

Question 26: (Option A) (a) Explain the events that constitute "double fertilization" in a typical angiosperm. (3 Marks) (b) What is the fate of the synergids and antipodal cells immediately following fertilization? (2 Marks) **OR (Option B)** (a) Discuss the life cycle of *Plasmodium vivax* in the human host, up to the point of

transmission back to the mosquito. (3 Marks) (b) Why do the symptoms of malaria (chills and high fever) occur in cyclic bursts? (2 Marks)

Question 27: (Option A) With the help of a well-labeled diagram, explain the typical structure of a biogas plant and the microbial processes involved in biogas production. (5 Marks) **OR (Option B)** Describe the major events of the menstrual cycle in human females, correlating the ovarian phases with changes in the levels of pituitary and ovarian hormones. (5 Marks)